#### 8.3.1.1.3 16-Bit CC

A 16-bit integrating ADC, commonly referred to as the coulomb counter (CC), provides measurements of accumulated charge across the current sense resistor. The integration period for this reading is 250 ms.

The CC may be operated in one of two modes: ALWAYS ON and 1-SHOT.

- In ALWAYS ON mode, the CC runs at 100%, gathering a fresh reading every 250 ms. The conclusion of each reading sets the CC_READY bit, which toggles the ALERT pin high to inform the microcontroller that a new reading is available. To enable Always On mode, set *[CC_EN]* = 1.
- In 1-SHOT mode, the CC performs a single 250-ms reading, and similarly sets the CC_READY bit when completed. This mode is intended for non-gauging usages, where the host simply desires to check the pack current.

  To enable a 1-SHOT reading, ensure *[CC_EN]* = 0 and set *[CC_ONESHOT]* = 1.

The fullscale range of the CC is ± 270 mV, with a max recommended input range of ± 200 mV, thus yielding an LSB of approximately 8.44 µV.

The following equation shows how to convert the 16-bit CC reading into an analog voltage if no board-level calibration is performed:

CC Reading (in µV) = ***[16-bit 2's Complement Value] × (8.44 µV/LSB)*** (3)

| 16-Bit CC Result | ADC Result in Decimal | CC Reading (in µV) |
|---|---|---|
| 0x0001 | 1 | 8.44 |
| 0x2710 | 10000 | 84,400 |
| 0x7D00 | 32000 | 270,080 |
| 0x8300 | –32000 | –270,080 |
| 0xC350 | –15536 | –131,123.84 |
| 0xFFFF | –1 | –8.44 |

#### 8.3.1.1.4 External Thermistor

One (BQ76920), two (BQ76930), or three (BQ76940) 10-kΩ NTC 103AT thermistors may be measured by the device. These are measured by applying a factory-trimmed internal 10-k pull-up resistance to an internal regulator value of nominally 3.3 V, the result of which can be read out from the TSx (TS1, TS2, TS3) registers.

To select thermistor measurement mode, set *[TEMP_SEL]* = 1.

Thermistor TS1 is connected between TS1 and VSS; TS2 is connected between TS2 and VC5x (BQ76930 and BQ76940 only); and TS3 is connected between TS3 and VC10x (BQ76940 only). These thermistors may be placed in various areas in the battery pack to measure such things as localized cell temperature, FET heating, and so forth.

The thermistor impedance may be calculated using the 14-bit ADC reading in the TS1, TS2, and TS3 registers and 10-k internal pull-up resistance as follows:

The following equations show how to use the 14-bit ADC readings in TS1, TS2, and TS3 to determine the resistance of the external 103AT thermistor:

$$V_{TSX} = (\text{ADC in Decimal}) \times 382\ \mu V/LSB \quad (4)$$

$$R_{TS} = (10{,}000 \times VTSX) \div (3.3 - VTSX) \quad (5)$$

To convert the thermistor resistance into temperature, please refer to the thermistor component manufacturer's data sheet.